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BS Biological Sciences Y4

Previous research indicated iron deficiency-induced chlorosis depended on phosphorus availability, which inhibited plants' photosynthesis. The experiment aims to investigate how the iron and phosphorus availability factors control plant's photosynthesis. The research results would contribute to revealing the effects of iron and phosphorus availability on plants gene regulation patterns, which is crucial to deeply understand the photosynthesis in plants. The 6 samples data is the RNA transcripts from the 7-day old *Arabidopsis thaliana* seedlings with treatment as phosphate and iron deprivation, which obtained by high-throughput sequencing technology. Particular, the samples are divided into control group (C1, C2, C3) and treatment group (T1, T2, T3). Arabidopsis wild-type plants in treatment group has an iron deficiency condition (-Fe+P), while control group has a normal condition (+Fe+P).

Data Quality Control and Alignment:

For quality control, the sample RNA were analyzed by fastqc functions and trimmed, the basic statistics of sample RNA before and after trimming were showed in Figure 1.A. In general, the control group RNA data (SRR13253052 file) and treatment group RNA data (SRR13253055 file) had sequences around 750000 bp with good qualities, the mean quality (blue line) of Per Base Sequence Quality was above 28 in untrimmed samples and over 32 in trimmed samples, which didn't include N content and adapters (Appendix 1 and Figure 1.B). However, the sequences included several overrepresented sequences and Kmer contents, and the mean GC deviations from the normal distribution were more than 15% (Figure 1.C), which might due to the small RNA libraries and biological reasons.

The trimmed sample fastq data was then aligned to the reference genome A.thaliana (TAIR 10) (version: release-52) to check the RNA reads location and RNA expression status by HISAT2 (Figure 1.D). All samples aligned the reference well with more than 93% overall alignment rates and more than 87% one-time alignment rates. While the overall alignment data was good, there were still multiple alignment sequences with percentage from 6% to 7%. Possible reasons might include homologous sequences in different genome locations, gene families, and incomplete reference. The sample alignment status was also viewed in IGV software as TDF file constructed from sorted BAM file. Most sample reads aligned to the A.thaliana (TAIR 10) chromosome 1 to 5 (Appendix.2).

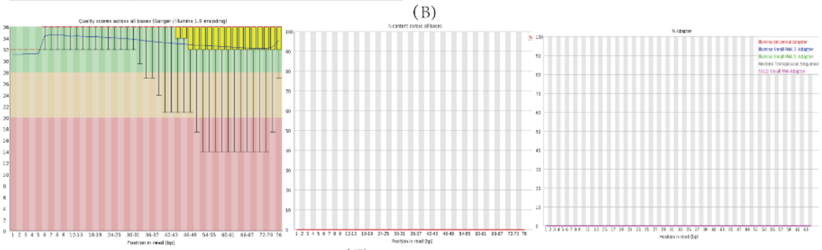
(A)

Samples	Total sequence	Sequence length	GC%	Sequences flagged as poor quality
C1	753088	76	44	0
C2	753088	76	44	0
C3	753088	76	44	0
T1	753088	76	44	0
T2	753088	76	44	0
T3	753088	76	44	0
TC1	753006	20-76	45	0
TC2	752993	20-76	44	0
TC3	753014	20-76	44	0
TT1	753021	20-76	44	0
TT2	753009	20-76	44	0
TT3	753014	20-76	44	0

(D)

Sample (total reads)	Overall alignment rate	Aligned 1 time	Aligned>1 times
C1 (753006)	96.26%	89.00%	7.27%
C2(752993)	95.56%	88.54%	7.01%
C3(753014)	94.79%	87.67%	7.12%
T1(753021)	96.04%	89.17%	6.87%
T2(753009)	94.97%	88.22%	6.76%
T3(753014)	95.22%	88.33%	6.88%

(B)



(C)

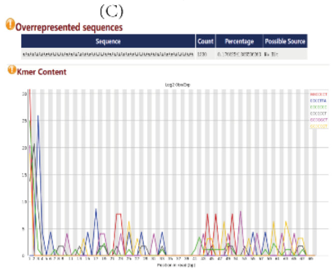


Figure 1. A. The basic RNA data of experiment samples before and after trimming. **B and C.** The high quality of trimmed RNA data and potential issues in the data (C1 as an example).
D. The alignment rate of trammed sample RNA data to the reference. Both control group and treatment group samples have high alignment rates to the A.thaliana (TAIR 10) reference.

Reference-based Transcriptome Reconstruction

To deeply evaluate the RNA data, the C1 sample were performed reference-based transcriptome reconstruction by SpringTie software (instead of Cufflinks) and visualized in IGV software. The transcript reads genome locations of re-assembled transcriptome was highly consistent to the A.thaliana (TAIR 10) reference transcripts (Figure 2.A), indicating the assembly was reconstructed as expected. Though the overall consistence of reconstructed transcriptome and reference transcriptome was high, the reconstructed transcriptome was different from the reference. In the region from 364,278-377,861 bp at chromosome 1 (Figure 2.B), the AT1G02070 and AT1G02074 reference transcripts were not appeared in the C1 sample, and in the C1 sample expressed reference transcripts AT1G02065.1 and AT1G02080.1, not all the exons were expressed in the corresponding C1 sample gene STRG.67.1, STRG.68.1, and STRG.70.1.

Furthermore, several novel transcripts were found in the C1 sample (Figure 2.C and D). In the region of 8,848,798- 8,850,002 bp and 9,589,594-9, 590,315 bp at chromosome 1, the appeared transcripts STRG.1346 and STRG.1437 in the C1 sample were not shown in the reference with low coverage value as 0.145709 and 0.068750 (less than 1) to the corresponding gene AT1G05943.1 and AT1G06103.1, respectively.

Figure 2.A. The reconstructed transcriptome highly aligned to the reference gene A.thaliana (TAIR 10), the assembly was working as expected. **B.** The reconstructed transcriptome was different from the reference, not all the reference genes and exons were expressed. **C and D.** The novel transcripts STRG.1346 and STRG.1437 found in reconstructed transcriptome chromosome 1.

Differential Gene Expression Analysis

To estimate the differential gene expression of *A.thaliana* under dissimilar experimental conditions, transcript reads of control and treatment samples were counted by featurecounts, analyzed by DEseq2 (RStudio), and visualized by ggplot2 (RStudio). At significance level of 0.05 (p-value<0.05), 183 genes were found expressed differently in the treatment groups to control groups, in which 53 genes were down-regulated and 130 genes up-regulated (Figure 3.A). Among them, gene AT4G25100 (the highest -log10 (padj) value in down-regulated gene) was re-checked by tag density files (TDF) in IGV (Figure 3.B). With same range (0-100), treatment group tag densities were significantly less than the control group, which was consistent to the differential analysis.

The differentially expressed genes were then performed gene ontology (GO) analysis in the DAVID website, for which the results were visualized by ggplot2 (RStudio). Figure 3.C and Figure 3.D illustrated the Top 4 significant (top 4 -log10 (P value)) genetic functions within the three category-Biological Process (BP), Cellular Component (CC), and Molecular Function (MF) in the upregulated and downregulated gene reads, respectively. Figure 3.E included a collection of significant ontology in BP, CC categories and KEGG functional pathways of up and down regulation genes. Based on the results, it could be concluded that in the treatment group with phosphorus but lack of iron, genes related to photosynthesis and chloroplasts of the plants were obviously down-regulated, while genes related to hypoxia and iron ion starvation were up-regulated. This meant that in the appearance of phosphorus but disappearance of iron, the plant's photosynthetic function would be weakened, and the plants activated some oxygen-related gene pathways to defend this dilemma.

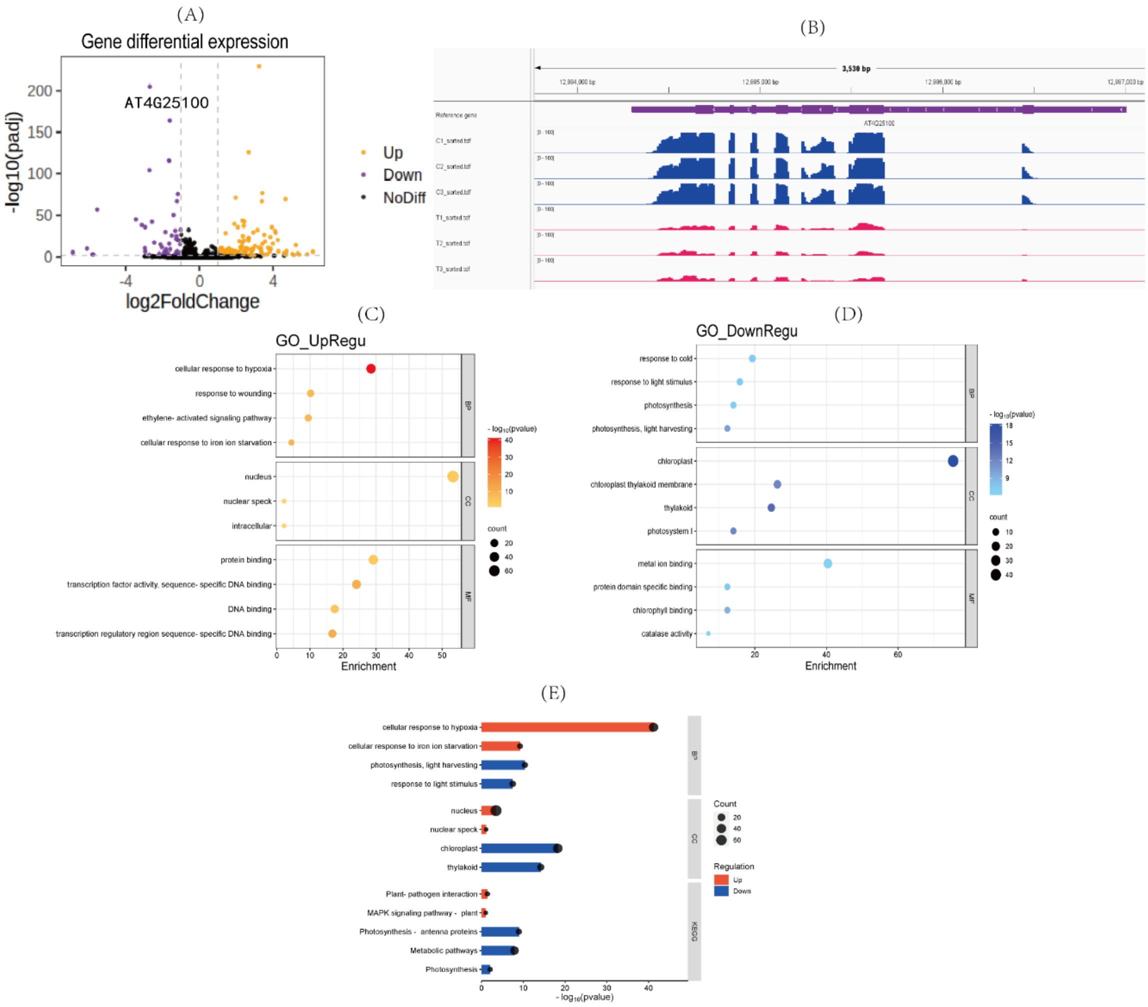
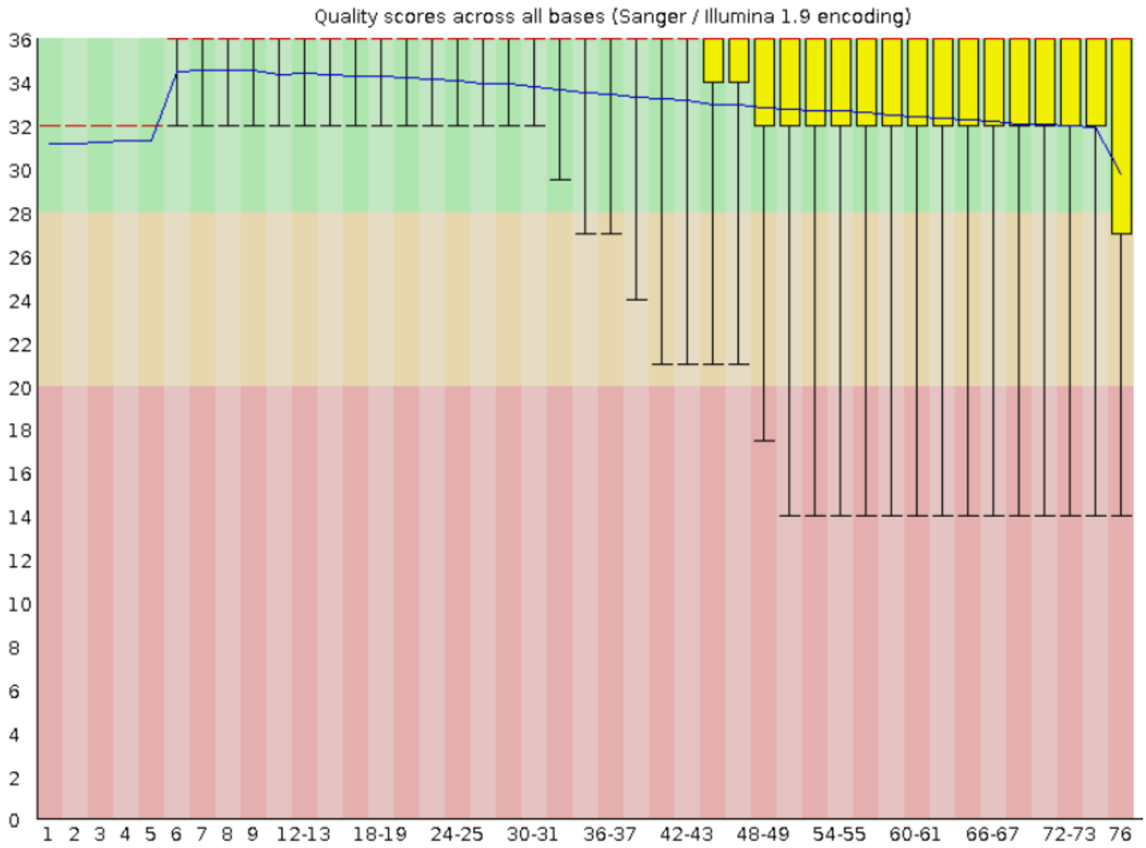


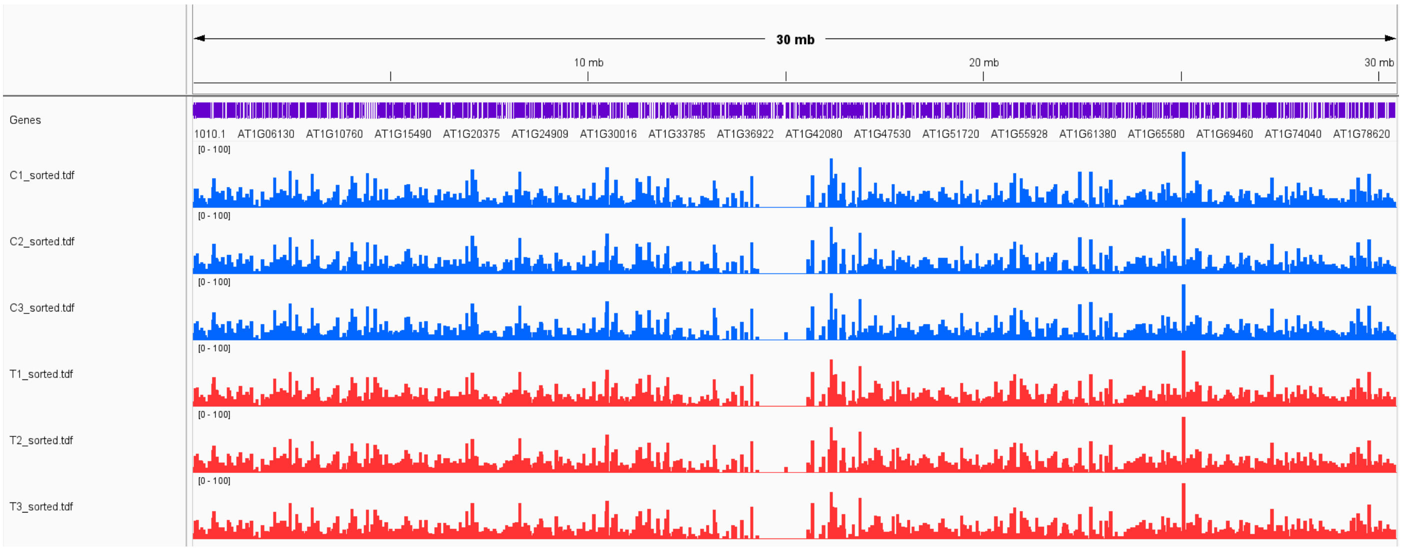
Figure 3. A. Visualization volcano of different gene expression level between treatment group and control group.
B. The TDF results of down-regulated gene AT4G25100 between the control samples and treatment samples were consistent to the differential analysis results. **C and D**. The GO analysis of up and downregulated genes. The enrichment was represented by gene percentage. Transcriptional factors and defending genes in upregulated gene and photosynthesis in downregulated gene were relatively significant. **E.** Selected representative GO and KEGG pathway comparison between Up and Down gene. Higher -log10(pvalue) indicated significant difference. In treatment group plants, photosynthesis related pathways were restricted while new transcriptional factor and MAPK signaling pathway were initiated.

Appendix

1. C1 as example (Before trimmed)



2. IGV visualization of sample alignment to reference gene



Important code annotation:

```
Fastqc:Check the data quality and purify the data
fastqc -o ~/p2/fastqc_report -f fastq \
/home/BIO316/P2/fastq/T1trimmed.fq

fastqc -o ~/p2/fastqc_report -f fastq \
/home/BIO316/P2/fastq/C1.fastq /home/BIO316/P2/fastq/C2.fastq \
/home/BIO316/P2/fastq/C3.fastq /home/BIO316/P2/fastq/T1.fastq \
/home/BIO316/P2/fastq/T2.fastq /home/BIO316/P2/fastq/T3.fastq

Aligh the data to reference in order to find reads location:
(nohup hisat2 -x /home/BIO316/P2/index/ -U ~/P2/fastq/C1_trimmed.fq -S \
~/P2/bam/C1.sam > ~/P2/bam/C1.out)

Convert SAM files to BAM files. Sort BAM files  (arrange in order) and build index (book list page) to increase the search rate


(nohup samtools view -S ~/P2/bam/C1.sam -b > ~/P2/bam/C1.bam)&

(nohup samtools sort ~/P2/bam/C1.bam -o ~/P2/bam/C1_sorted.bam)&
(nohup samtools index ~/P2/bam/C1_sorted.bam)&

Convert to TDF(signal of BAM) which could be checked in IGV

(nohup igvtools count -z 5 -w 10 -e 0 ~/P2/bam/C1_sorted.bam \
~/P2/tdf/C1_sorted.tdf /home/BIO316/P2/TAIR10.chrom.sizes)&

Reference-based reconstruction transcriptome
(nohup stringtie -G /home/BIO316/P2/index/TAIR10_52.gtf -o ~/P2/GTF/C1.gtf \
~/P2/bam/C1_sorted.bam -A ~/P2/GTF/C1.txt)&

Gene expression estimation by featurecounts (count the reads in each gene location)
(nohup featureCounts -t exon -g gene_id \
-a /home/BIO316/P2/index/TAIR10_52.gtf -o ~/P2/Exp/feature.txt \
~/P2/bam/C1_sorted.bam ~/P2/bam/C2_sorted.bam ~/P2/bam/C3_sorted.bam \
~/P2/bam/T1_sorted.bam ~/P2/bam/T2_sorted.bam ~/P2/bam/T3_sorted.bam \
> ~/P2/featureCounts.out)&
```